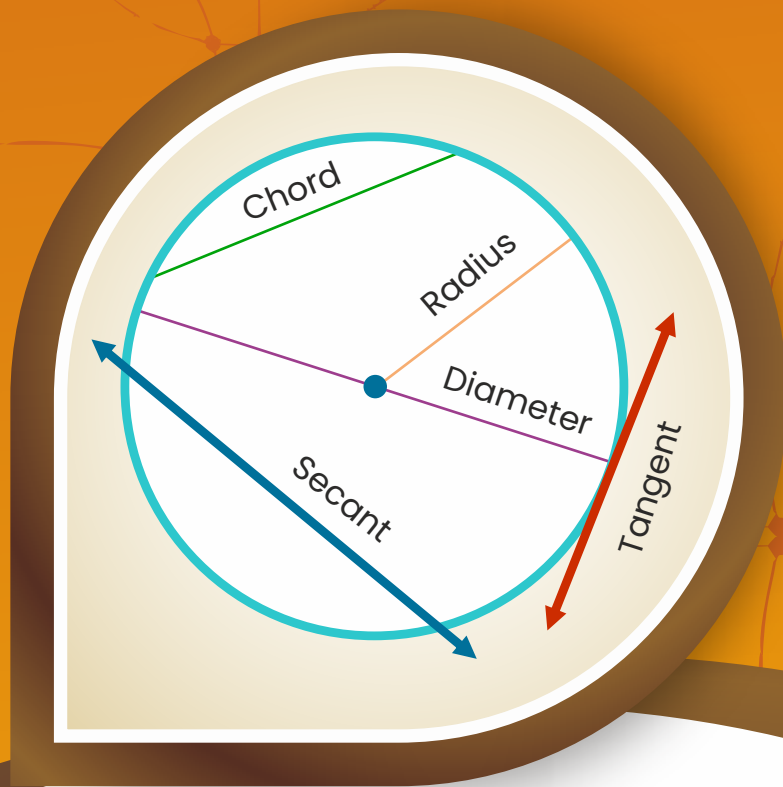


MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Circle

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. Centres of the three circles $x^2 + y^2 - 4x - 6y - 14 = 0$, $x^2 + y^2 + 2x + 4y - 5 = 0$ and $x^2 + y^2 - 10x - 16y + 7 = 0$
 (A) are the vertices of a right triangle
 (B) the vertices of an isosceles triangle which is not regular
 (C) vertices of a regular triangle
 (D) are collinear

CR0001

2. $y - 1 = m_1(x - 3)$ and $y - 3 = m_2(x - 1)$ are two family of straight lines, at right angled to each other. The locus of their point of intersection is
 (A) $x^2 + y^2 - 2x - 6y + 10 = 0$ (B) $x^2 + y^2 - 4x - 4y + 6 = 0$
 (C) $x^2 + y^2 - 2x - 6y + 6 = 0$ (D) $x^2 + y^2 - 4x - 4y - 6 = 0$

CR0002

3. Suppose that the equation of the circle having $(-3, 5)$ and $(5, -1)$ as end points of a diameter is $(x - a)^2 + (y - b)^2 = r^2$. Then $a + b + r$, ($r > 0$) is
 (A) 8 (B) 9 (C) 10 (D) 11

CR0003

4. The area of an equilateral triangle inscribed in the circle $x^2 + y^2 - 2x = 0$ is
 (A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3\sqrt{3}}{8}$ (D) none

CR0004

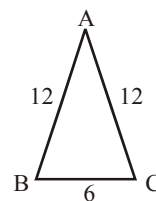
5. The smallest distance between the circle $(x - 5)^2 + (y + 3)^2 = 1$ and the line $5x + 12y - 4 = 0$, is
 (A) $1/13$ (B) $2/13$ (C) $3/15$ (D) $4/15$

CR0005

6. The equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$ is
 (A) $x^2 + y^2 + 32x - 4y + 235 = 0$ (B) $x^2 + y^2 + 32x + 4y - 235 = 0$
 (C) $x^2 + y^2 + 32x - 4y - 235 = 0$ (D) $x^2 + y^2 + 32x + 4y + 235 = 0$

CR0006

7. The radius of the circle passing through the vertices of the triangle ABC, is
 (A) $\frac{8\sqrt{15}}{5}$ (B) $\frac{3\sqrt{15}}{5}$
 (C) $3\sqrt{5}$ (D) $3\sqrt{2}$



CR0007

8. (6, 0), (0, 6) and (7, 7) are the vertices of a triangle. The circle inscribed in the triangle has the equation

(A) $x^2 + y^2 - 9x + 9y + 36 = 0$

(B) $x^2 + y^2 - 9x - 9y + 36 = 0$

(C) $x^2 + y^2 + 9x - 9y + 36 = 0$

(D) $x^2 + y^2 - 9x - 9y - 36 = 0$

CR0008

9. The line joining (5, 0) to $(10\cos\theta, 10\sin\theta)$ is divided internally in the ratio 2 : 3 at P. If θ varies then the locus of P is :

(A) a pair of straight lines

(B) a circle

(C) a straight line

(D) a second degree curve which is not a circle

CR0009

10. The locus of the center of the circles such that the point (2, 3) is the mid point of the chord $5x + 2y = 16$ is

(A) $2x - 5y + 11 = 0$

(B) $2x + 5y - 11 = 0$

(C) $2x + 5y + 11 = 0$

(D) none

CR0010

11. In the xy-plane, the length of the shortest path from (0, 0) to (12, 16) that does not go inside the circle $(x - 6)^2 + (y - 8)^2 = 25$ is

(A) $10\sqrt{3}$

(B) $10\sqrt{5}$

(C) $10\sqrt{3} + \frac{5\pi}{3}$

(D) $10 + 5\pi$

CR0011

12. Tangents PA and PB are drawn to the circle $x^2 + y^2 = 4$, then the locus of the point P if the triangle PAB is equilateral, is equal to-

(A) $x^2 + y^2 = 16$

(B) $x^2 + y^2 = 8$

(C) $x^2 + y^2 = 64$

(D) $x^2 + y^2 = 32$

CR0012

13. The points (x_1, y_1) , (x_2, y_2) , (x_1, y_2) and (x_2, y_1) are always

(A) collinear

(B) concyclic

(C) vertices of a square

(D) vertices of a rhombus

CR0013

14. Locus of all point P(x, y) satisfying $x^3 + y^3 + 3xy = 1$ consists of union of

(A) a line and an isolated point

(B) a line pair and an isolated point

(C) a line and a circle

(D) a circle and a isolated point.

CR0014

15. In the xy plane, the segment with end points (3, 8) and (-5, 2) is the diameter of the circle. The point (k, 10) lies on the circle for

(A) no value of k

(B) exactly one integral k

(C) exactly one non integral k

(D) two real values of k

CR0015

- 16.** If a circle of constant radius $3k$ passes through the origin 'O' and meets co-ordinate axes at A and B then the locus of the centroid of the triangle OAB is -

(A) $x^2 + y^2 = (2k)^2$ (B) $x^2 + y^2 = (3k)^2$ (C) $x^2 + y^2 = (4k)^2$ (D) $x^2 + y^2 = (6k)^2$

CR0016

- 17.** Consider the points P(2, 1); Q(0, 0); R(4, -3) and the circle S : $x^2 + y^2 - 5x + 2y - 5 = 0$

(A) exactly one point lies outside S (B) exactly two points lie outside S
(C) all the three points lie outside S (D) none of the point lies outside S

CR0017

- 18.** B and C are fixed points having co-ordinates $(3, 0)$ and $(-3, 0)$ respectively. If the vertical angle BAC is 90° , then the locus of the centroid of the $\triangle ABC$ has the equation :

(A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 = 2$ (C) $9(x^2 + y^2) = 1$ (D) $9(x^2 + y^2) = 4$

CR0018

- 19.** The angle between the two tangents from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ equals

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

CR0019

- 20.** Tangents are drawn from $(4, 4)$ to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ to meet the circle at A and B. The length of the chord AB is

(A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $2\sqrt{6}$ (D) $6\sqrt{2}$

CR0020

- 21.** The area of the quadrilateral formed by the tangents from the point $(4, 5)$ to the circle $x^2 + y^2 - 4x - 2y - 11 = 0$ with the pair of radii through the points of contact of the tangents is :

(A) 4 sq. units (B) 8 sq. units (C) 6 sq. units (D) none

CR0021

22. If L_1 and L_2 are the length of the tangent from $(0, 5)$ to the circles $x^2 + y^2 + 2x - 4 = 0$ and $x^2 + y^2 - 2x - y + 1 = 0$ then

(A) $L_1 = 2L_2$ (B) $L_2 = 2L_1$ (C) $L_1 = L_2$ (D) $L_1^2 = L_2$

CR0022

- 23.** From (3, 4) chords are drawn to the circle $x^2 + y^2 - 4x = 0$. The locus of the mid points of the chords is :

(A) $x^2 + y^2 - 5x - 4y + 6 = 0$ (B) $x^2 + y^2 + 5x - 4y + 6 = 0$
(C) $x^2 + y^2 - 5x + 4y + 6 = 0$ (D) $x^2 + y^2 - 5x - 4y - 6 = 0$

CR0023

24. Tangents are drawn to a unit circle with centre at the origin from each point on the line $2x + y = 4$. Then the equation to the locus of the middle point of the chord of contact is -

(A) $2(x^2 + y^2) = x + y$ (B) $2(x^2 + y^2) = x + 2y$ (C) $4(x^2 + y^2) = 2x + y$ (D) none

CR0024

25. Chord AB of the circle $x^2 + y^2 = 100$ passes through the point (7, 1) and subtends an angle of 60° at the circumference of the circle. If m_1 and m_2 are the slopes of two such chords then the value of $m_1 m_2$, is

(A) -1 (B) 1 (C) $7/12$ (D) -3

CR0025

26. Combined equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 4x + 6y + 9 = 0$ is

(A) $3(x^2 + y^2) = (x + 2y)^2$ (B) $2(x^2 + y^2) = (3x + y)^2$
(C) $9(x^2 + y^2) = (2x + 3y)^2$ (D) $x^2 + y^2 = (2x + 3y)^2$

CR0026

27. Sum of the abscissa and ordinate of the centre of the circle touching the line $3x + y + 2 = 0$ at the point (-1, 1) and passing through the point (3, 5) is-

(A) 2 (B) 3 (C) 4 (D) 5

CR0027

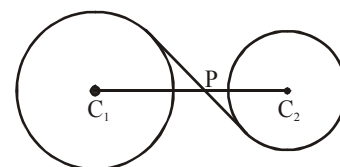
28. A circle of radius 5 is tangent to the line $4x - 3y = 18$ at M(3, -2) and lies above the line. The equation of the circle, is-

(A) $x^2 + y^2 - 6x + 4y - 12 = 0$ (B) $x^2 + y^2 + 2x - 2y - 3 = 0$
(C) $x^2 + y^2 + 2x - 2y - 23 = 0$ (D) $x^2 + y^2 + 6x + 4y - 12 = 0$

CR0028

29. In the figure given, two circles with centres C_1 and C_2 are 35 units apart, i.e. $C_1 C_2 = 35$. The radii of the circles with centres C_1 and C_2 are 12 and 9 respectively. If P is the intersection of $C_1 C_2$ and a common internal tangent to the circles, then $l(C_1 P)$ equals-

(A) 18 (B) 20 (C) 12 (D) 15



CR0029

30. Let C_1 and C_2 are circles defined by $x^2 + y^2 - 20x + 64 = 0$ and $x^2 + y^2 + 30x + 144 = 0$. The length of the shortest line segment PQ that is tangent to C_1 at P and to C_2 at Q is -

(A) 15 (B) 18 (C) 20 (D) 24

CR0030

EXERCISE (O-2)

Multiple Correct Answer Type

1. $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$, represents : (Where x_1, y_1 are constant)

(A) equation of a straight line, if θ is constant and r is variable
 (B) equation of a circle, if r is constant and θ is a variable
 (C) a straight line passing through a fixed point and having a known slope
 (D) a circle with a known centre and a given radius.

CR0039

2. Which of the following lines have the intercepts of equal lengths on the circle, $x^2 + y^2 - 2x + 4y = 0$?

(A) $3x - y = 0$ (B) $x + 3y = 0$
 (C) $x + 3y + 10 = 0$ (D) $3x - y - 10 = 0$

CR0038

3. Consider two circles $C_1 : x^2 + y^2 - 1 = 0$ and $C_2 : x^2 + y^2 - 2 = 0$. Let $A(1, 0)$ be a fixed point on the circle C_1 and B be any variable point on the circle C_2 . The line BA meets the curve C_2 again at C . Which of the following alternative(s) is/are correct ?

(A) $OA^2 + OB^2 + BC^2 \in [7, 11]$, where O is the origin.
 (B) $OA^2 + OB^2 + BC^2 \in [4, 7]$, where O is the origin.

(C) Locus of midpoint of AB is a circle of radius $\frac{1}{\sqrt{2}}$.

(D) Locus of midpoint of AB is a circle of area $\frac{\pi}{2}$.

CR0163

4. One of the diameter of the circle circumscribing the rectangle $ABCD$ is $x - 3y + 1 = 0$.

If two vertices of rectangle are the points $(-2, 5)$ and $(6, 5)$ respectively, then which of the following hold(s) good?

(A) Area of rectangle $ABCD$ is 64 square units.
 (B) Centre of circle is $(2, 1)$
 (C) The other two vertices of the rectangle are $(-2, -3)$ and $(6, -3)$
 (D) Equation of sides are $x = -2$, $y = -3$, $x = 5$ and $y = 6$.

CR0164

5. Three concentric circles of which the biggest is $x^2 + y^2 = 1$, have their radii in A.P. If the line $y = x + 1$ cuts all the circles in real and distinct points. The permissible values of common difference of A.P. is/are

(A) 0.4 (B) 0.6 (C) 0.01 (D) 0.1

R0165

6. If $4\ell^2 - 5m^2 + 6\ell + 1 = 0$, then line $\ell x + my + 1 = 0$ touches a definite circle, then which of the following is/are true.

(A) Centre (0, 3) (B) centre (3, 0) (C) Radius $\sqrt{5}$ (D) Radius 5

CR0166

7. The equation of circles passing through (3, -6) touching both the axes is

(A) $x^2 + y^2 - 6x + 6y + 9 = 0$ (B) $x^2 + y^2 + 6x - 6y + 9 = 0$
(C) $x^2 + y^2 + 30x - 30y + 225 = 0$ (D) $x^2 + y^2 - 30x + 30y + 225 = 0$

CR0167

8. Tangents are drawn to the circle $x^2 + y^2 = 50$ from a point 'P' lying on the x-axis. These tangents meet the y-axis at points 'P₁' and 'P₂'. Possible coordinates of 'P' so that area of triangle PP₁P₂ is minimum, is/are

(A) (10, 0) (B) $(10\sqrt{2}, 0)$ (C) (-10, 0) (D) $(-10\sqrt{2}, 0)$

CR0168

9. $x^2 + y^2 = a^2$ and $(x - 2a)^2 + y^2 = a^2$ are two equal circles touching each other. Find the equation of circle (or circles) of the same radius touching both the circles

(A) $x^2 + y^2 + 2ax + 2\sqrt{3}ay + 3a^2 = 0$ (B) $x^2 + y^2 - 2ax + 2\sqrt{3}ay + 3a^2 = 0$
(C) $x^2 + y^2 + 2ax - 2\sqrt{3}ay + 3a^2 = 0$ (D) $x^2 + y^2 - 2ax - 2\sqrt{3}ay + 3a^2 = 0$

CR0169

10. A family of linear functions is given by $f(x) = 1 + c(x + 3)$ where $c \in \mathbb{R}$. If a member of this family meets a unit circle centred at origin in two coincident points then 'c' can be equal to

(A) -3/4 (B) 0 (C) 3/4 (D) 1

CR0040

11. The equations of the tangents drawn from the origin to the circle, $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are :

(A) $x = 0$ (B) $y = 0$
(C) $(h^2 - r^2)x - 2rhy = 0$ (D) $(h^2 - r^2)x + 2rhy = 0$

CR0041

12. Tangents PA and PB are drawn to the circle $S \equiv x^2 + y^2 - 2y - 3 = 0$ from the point P(3,4). Which of the following alternative(s) is/are correct ?

(A) The power of point P(3,4) with respect to circle $S = 0$ is 14.
(B) The angle between tangents from P(3,4) to the circle $S = 0$ is $\frac{\pi}{3}$
(C) The equation of circumcircle of ΔPAB is $x^2 + y^2 - 3x - 5y + 4 = 0$
(D) The area of quadrilateral PACB is $3\sqrt{7}$ square units where C is the centre of circle $S = 0$.

CR0042

13. Consider the circles $C_1 : x^2 + y^2 = 16$ and $C_2 : x^2 + y^2 - 12x + 32 = 0$. Which of the following statement is/are correct ?
- (A) Number of common tangent to these circles is 3.
 (B) The point P with coordinates (4,1) lies outside the circle C_1 and inside the circle C_2 .
 (C) Their direct common tangent intersect at (12,0).
 (D) Slope of their radical axis is not defined.

CR0043

14. Which of the following is/are True ?
 The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that -
- (A) they do not intersect
 (B) they touch each other
 (C) their exterior common tangents are parallel.
 (D) their interior common tangents are perpendicular.

CR0044

15. Consider the circles $S_1 : x^2 + y^2 = 4$ and $S_2 : x^2 + y^2 - 2x - 4y + 4 = 0$ which of the following statements are correct ?
- (A) Number of common tangents to these circles is 2.
 (B) If the power of a variable point P w.r.t. these two circles is same then P moves on the line $x + 2y - 4 = 0$
 (C) Sum of the y-intercepts of both the circles is 6.
 (D) The circles S_1 and S_2 are orthogonal.

CR0045

16. For the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ which of the following is/are true.
- (A) Number of integral values of r are 14 for which circles are intersecting.
 (B) Number of integral values of r are 9 for which circles are intersecting.
 (C) For r equal to 13 number of common tangents are 3.
 (D) For r equal to 21 number of common tangents are 2.

CR0170

17. Which of the following statement(s) is/are correct with respect to the circles $S_1 \equiv x^2 + y^2 - 4 = 0$ and $S_2 \equiv x^2 + y^2 - 2x - 4y + 4 = 0$?
- (A) S_1 and S_2 intersect at an angle of 90° .
 (B) The point of intersection of the two circle are (2, 0) and $\left(\frac{6}{5}, \frac{8}{5}\right)$.
 (C) Length of the common of chord of S_1 and S_2 is $\frac{4}{\sqrt{5}}$.
 (D) The point (2, 3) lies outside the circles S_1 and S_2 .

CR0171

18. Two circles of radii r_1 and r_2 are both touching the coordinate axes and intersecting each other orthogonally. The value of r_1/r_2 equals -

(A) $2 + \sqrt{3}$ (B) $\sqrt{3} + 1$ (C) $2 - \sqrt{3}$ (D) $2 + \sqrt{5}$

CR0035

19. Two circles $x^2 + y^2 + px + py - 7 = 0$ and $x^2 + y^2 - 10x + 2py + 1 = 0$ intersect each other orthogonally then the value of p is -

(A) 1 (B) 2 (C) 3 (D) 5

CR0046

20. If the circle $C_1: x^2 + y^2 = 16$ intersects another circle C_2 of radius 5 in such a manner that the common chord is of maximum length and has a slope equal to $3/4$, then the co ordinates of the centre of C_2 are:

(A) $\left(\frac{9}{5}, \frac{12}{5}\right)$ (B) $\left(\frac{9}{5}, -\frac{12}{5}\right)$ (C) $\left(-\frac{9}{5}, -\frac{12}{5}\right)$ (D) $\left(-\frac{9}{5}, +\frac{12}{5}\right)$

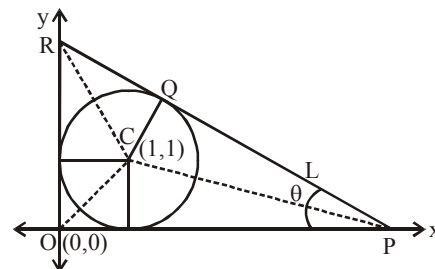
CR0172

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question nos. 1 to 3

In the diagram as shown, a circle is drawn with centre $C(1, 1)$ and radius 1 and a line L . The line L is tangential to the circle at Q . Further L meet the y -axis at R and the x -axis at P is such a way that the angle OPQ equals θ where $0 < \theta < \frac{\pi}{2}$.



- The coordinates of Q are
 (A) $(1 + \cos\theta, 1 + \sin\theta)$ (B) $(\sin\theta, \cos\theta)$
 (C) $(1 + \sin\theta, \cos\theta)$ (D) $(1 + \sin\theta, 1 + \cos\theta)$
- Equation of the line PR is
 (A) $x\cos\theta + y\sin\theta = \sin\theta + \cos\theta + 1$ (B) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta - 1$
 (C) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta + 1$ (D) $x \tan \theta + y = 1 + \cot\left(\frac{\theta}{2}\right)$

CR0047

- If the area bounded by the circle, the x -axis and PQ is $A(\theta)$, then $A\left(\frac{\pi}{4}\right)$ equals
 (A) $\sqrt{2} + 1 - \frac{3\pi}{8}$ (B) $\sqrt{2} - 1 + \frac{3\pi}{8}$ (C) $\sqrt{2} + 1 + \frac{\pi}{8}$ (D) $\sqrt{2} - 1 + \frac{\pi}{8}$

CR0047

Paragraph for question Nos. 4 and 5

Consider the circle $S : x^2 + y^2 - 4x - 1 = 0$ and the line $L : y = 3x - 1$. If the line L cuts the circle at A & B .

- Which of the following option(s) is/are **CORRECT** ?
 (A) Length of the chord AB equal is $\sqrt{10}$ unit
 (B) The angle subtended by the chord AB in the minor arc of S is $\frac{3\pi}{4}$
 (C) Acute angle between the line L and the circle S is $\frac{\pi}{4}$
 (D) Area bounded by $L = 0$ and minor arc of $S = 0$ is $\frac{5}{4}(\pi - 2)$ sq. unit

CR0173

5. If the equation of the circle on AB as diameter is of the form $x^2 + y^2 + ax + by + c = 0$ then

(A) magnitude of the vector $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$ has the value equal to $\sqrt{6}$

(B) $|a + b + c| = 4$ (C) $|a| + |b| + |c| = 5$ (D) $a^3 + b^3 + c^3 = 10$

CR0174

Paragraph for question Nos. 6 to 8

Let S_1, S_2, S_3 be the circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$, respectively then

6. Point from which length of tangents to these three circles is same is (α, β) then $(\alpha + \beta)$ is

CR0175

7. Equation of circle S_4 which cut orthogonally to all given circle is $x^2 + y^2 + ax + by + c = 0$ then $|a + b + c|$ is

CR0176

8. Radical centre of circles S_1, S_2 & S_4 is (p, q) then $|p + q|$ is

CR0177

9. Consider two circles C_1 of radius 'a' and C_2 of radius 'b' ($b > a$) both lying in the first quadrant and touching the coordinate axes. In each of the conditions listed in **List-I**, the ratio of b/a is given in **List-II**.

List-I

List-II

(I) C_1 and C_2 touch each other

(P) $2 + \sqrt{2}$

(II) C_1 and C_2 are orthogonal

(Q) 3

(III) C_1 and C_2 intersect so that the common chord is longest

(R) $2 + \sqrt{3}$

(IV) C_2 passes through the centre of C_1

(S) $3 + 2\sqrt{2}$

(T) $3 - 2\sqrt{2}$

(A) (I) $\rightarrow$ (S), (II) $\rightarrow$ (R), (III) $\rightarrow$ (Q), (IV) $\rightarrow$ (P)

(B) (I) $\rightarrow$ (S), (II) $\rightarrow$ (R), (III) $\rightarrow$ (T), (IV) $\rightarrow$ (Q)

(C) (I) $\rightarrow$ (R), (II) $\rightarrow$ (T), (III) $\rightarrow$ (Q), (IV) $\rightarrow$ (P)

(D) (I) $\rightarrow$ (P), (II) $\rightarrow$ (R), (III) $\rightarrow$ (S), (IV) $\rightarrow$ (T)

CR0114

10. **Column - I**

Column - II

(A) If director circle of two given circles C_1 and C_2 of equal radii touches each other, then ratio of length of internal common tangent of C_1 and C_2 to their radii equals to

(P) 13

(B) Let two circles having radii r_1 and r_2 are orthogonal to each other. If length of their common chord is k times the square root of harmonic mean between squares of their radii, then k^4 equals to

(Q) 7

(C) The axes are translated so that the new equation of the circle $x^2 + y^2 - 5x + 2y - 5 = 0$ has no first degree terms and the new equation $x^2 + y^2 = \frac{\lambda^2}{4}$, then a value of λ is

(R) 4

(D) The number of integral points which lie on or inside the circle $x^2 + y^2 = 4$ is

(S) 2

CR0178

EXERCISE (O-4)**Numerical Grid Type**

1. A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y -axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$ where $\sqrt{a}$ is a surd, find the value of $a + b$.
CR0105
2. If the points $(\lambda, -\lambda)$ lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$, then number of integers in the range of λ is
CR0068
3. One of the diameters of the circle circumscribing the rectangle ABCD is $4y = x + 7$. If A & B are the points $(-3, 4)$ & $(5, 4)$ respectively, then find the area of the rectangle. (in square unit)
CR0058
4. Circles C_1 and C_2 are externally tangent and they are both internally tangent to the circle C_3 . The radii of C_1 and C_2 are 4 and 10, respectively and the centres of the three circles are collinear. A chord of C_3 is also a common internal tangent of C_1 and C_2 . Given that the length of the chord is $\frac{m\sqrt{n}}{p}$ where m , n and p are positive integers, m and p are relatively prime and n is not divisible by the square of any prime, find the value of $(m + n + p)$.
CR0083
5. Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres $(-3, 0)$, $(-1, 0)$, $(1, 0)$ and $(3, 0)$ respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C. If the length of this chord can be expressed as $\sqrt{x}$, find x .
CR0084
6. A line with gradient 2 is passing through the point $P(1, 7)$ and touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at the point Q. If (a, b) are the coordinates of the point Q, then find the value of $(7a + 7b + c)$.
CR0079
7. Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression $z = \frac{4-y}{7-x}$ are M and m respectively, then find the value $(2M + 6m)$.
CR0103
8. Let $S_1 = 0$ and $S_2 = 0$ be two circles intersecting at $P(6, 4)$ and both are tangent to x -axis and line $y = mx$ (where $m > 0$). If product of radii of the circles $S_1 = 0$ and $S_2 = 0$ is $\frac{52}{3}$, then the value of m^2 is
CR0106

9. Through a given point $P(5, 2)$, secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$,

$PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$. Find the value of $\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2$.

[Note : $A_r(B_r)$ denotes that the line passing through $P(5, 2)$ meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r .]

CR0107

10. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where $a, b > 0$), then find the maximum value of (ab) .

CR0102

EXERCISE (JM)

1. The number of common tangents to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$, is : [JEE(Main)-2015]
(1) 3 (2) 4 (3) 1 (4) 2
CR0125
2. If one of the diameters of the circle, given by the equation, $x^2 + y^2 - 4x + 6y - 12 = 0$, is a chord of a circle S, whose centre is at $(-3, 2)$, then the radius of S is :- [JEE(Main)-2016]
(1) 10 (2) $5\sqrt{2}$ (3) $5\sqrt{3}$ (4) 5
CR0126
3. The centres of those circles which touch the circle, $x^2 + y^2 - 8x - 8y - 4 = 0$, externally and also touch the x-axis, lie on :- [JEE(Main)-2016]
(1) A parabola (2) A circle
(3) An ellipse which is not a circle (4) A hyperbola
CR0127
4. Three circles of radii a, b, c ($a < b < c$) touch each other externally. If they have x-axis as a common tangent, then : [JEE(Main)-2019]
(1) $\frac{1}{\sqrt{a}} = \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}}$ (2) a, b, c are in A.P.
(3) $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in A.P. (4) $\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}}$
CR0128
5. If a circle C passing through the point $(4, 0)$ touches the circle $x^2 + y^2 + 4x - 6y = 12$ externally at the point $(1, -1)$, then the radius of C is : [JEE(Main)-2019]
(1) $\sqrt{57}$ (2) 4 (3) $2\sqrt{5}$ (4) 5
CR0129
6. If the area of an equilateral triangle inscribed in the circle, $x^2 + y^2 + 10x + 12y + c = 0$ is $27\sqrt{3}$ sq. units then c is equal to : [JEE(Main)-2019]
(1) 20 (2) 25 (3) 13 (4) -25
CR0130
7. A square is inscribed in the circle $x^2 + y^2 - 6x + 8y - 103 = 0$ with its sides parallel to the coordinate axes. Then the distance of the vertex of this square which is nearest to the origin is :- [JEE(Main)-2019]
(1) 13 (2) $\sqrt{137}$ (3) 6 (4) $\sqrt{41}$
CR0131
8. A circle cuts a chord of length $4a$ on the x-axis and passes through a point on the y-axis, distant $2b$ from the origin. Then the locus of the centre of this circle, is :- [JEE(Main)-2019]
(1) A hyperbola (2) A parabola (3) A straight line (4) An ellipse

9. If a variable line, $3x + 4y - \lambda = 0$ is such that the two circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ are on its opposite sides, then the set of all values of λ is the interval :-

[JEE(Main)-2019]

- (1) [12, 21] (2) (2, 17) (3) (23, 31) (4) [13, 23]

CR0133

10. The sum of the squares of the lengths of the chords intercepted on the circle, $x^2 + y^2 = 16$, by the lines, $x + y = n$, $n \in \mathbb{N}$, where $\mathbb{N}$ is the set of all natural numbers, is :

[JEE(Main)-2019]

- (1) 320 (2) 160 (3) 105 (4) 210

CR0134

11. If a tangent to the circle $x^2 + y^2 = 1$ intersects the coordinate axes at distinct points P and Q, then the locus of the mid-point of PQ is

[JEE(Main)-2019]

- (1) $x^2 + y^2 - 2xy = 0$ (2) $x^2 + y^2 - 16x^2y^2 = 0$
(3) $x^2 + y^2 - 4x^2y^2 = 0$ (4) $x^2 + y^2 - 2x^2y^2 = 0$

CR0135

12. The common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 6x + 8y - 24 = 0$ also passes through the point :-

[JEE(Main)-2019]

- (1) (-4, 6) (2) (6, -2) (3) (-6, 4) (4) (4, -2)

CR0136

13. Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B. The $(AB)^2$ is equal to :

[JEE(Main)-2020]

- (1) $\frac{52}{5}$ (2) $\frac{32}{5}$ (3) $\frac{56}{5}$ (4) $\frac{64}{5}$

CR0137

14. If a line, $y = mx + c$ is a tangent to the circle, $(x - 3)^2 + y^2 = 1$ and it is perpendicular to a line L_1 ,

where L_1 is the tangent to the circle, $x^2 + y^2 = 1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, then [JEE(Main)-2020]

- (1) $c^2 - 6c + 7 = 0$ (2) $c^2 + 6c + 7 = 0$ (3) $c^2 + 7c + 6 = 0$ (4) $c^2 - 7c + 6 = 0$

CR0138

15. If the curves, $x^2 - 6x + y^2 + 8 = 0$ and $x^2 - 8y + y^2 + 16 - k = 0$, ($k > 0$) touch each other at a point, then the largest value of k is _____.

[JEE(Main)-2020]

CR0139

16. If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord of another circle 'C', whose center is at (2, 1), then its radius is _____.

[JEE(Main)-2021]

CR0179

17. If the locus of the mid-point of the line segment from the point (3, 2) to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r , then r is equal to :

[JEE(Main)-2021]

- (1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) $\frac{1}{4}$

CR0180

18. Let the normals at all the points on a given curve pass through a fixed point (a, b) . If the curve passes through $(3, -3)$ and $(4, -2\sqrt{2})$, and given that $a - 2\sqrt{2}b = 3$, then $(a^2 + b^2 + ab)$ is equal to _____.

[JEE(Main)-2021]

CR0181

19. Let the lengths of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$, $(a < 0)$ be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to :

[JEE(Main)-2021]

- (1) $\sqrt{11}$ (2) $\sqrt{7}$ (3) $\sqrt{6}$ (4) $\sqrt{10}$

CR0182

20. The minimum distance between any two points P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations
 $x^2 + y^2 - 10x - 10y + 41 = 0$
 $x^2 + y^2 - 24x - 10y + 160 = 0$ is _____.

[JEE(Main)-2021]

CR0183

21. Let a circle $C : (x - h)^2 + (y - k)^2 = r^2$, $k > 0$, touch the x-axis at $(1, 0)$. If the line $x + y = 0$ intersects the circle C at P and Q such that the length of the chord PQ is 2, then the value of $h + k + r$ is equal to _____.

[JEE(Main)-2022]

CR0184

22. The set of values of k for which the circle $C : 4x^2 + 4y^2 - 12x + 8y + k = 0$ lies inside the fourth quadrant and the point $\left(1, -\frac{1}{3}\right)$ lies on or inside the circle C is :

[JEE(Main)-2022]

- (A) An empty set (B) $\left[6, \frac{95}{9}\right]$ (C) $\left[\frac{80}{9}, 10\right]$ (D) $\left[9, \frac{92}{9}\right]$

CR0185

23. A rectangle R with end points of the one of its sides as $(1, 2)$ and $(3, 6)$ is inscribed in a circle. If the equation of a diameter of the circle is $2x - y + 4 = 0$, then the area of R is _____.

[JEE(Main)-2022]

CR0186

24. Let the tangent to the circle $C_1 : x^2 + y^2 = 2$ at the point $M(-1, 1)$ intersect the circle $C_2 : (x - 3)^2 + (y - 2)^2 = 5$, at two distinct points A and B . If the tangents to C_2 at the points A and B intersect at N , then the area of the triangle ANB is equal to :

[JEE(Main)-2022]

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) $\frac{5}{3}$

CR0187

25. Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is

[JEE(Main)-2022]

- (A) 12 (B) 13 (C) 14 (D) 16

CR0188

EXERCISE (JA)

1. Circle(s) touching x-axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ or y-axis is (are) [JEE(Advanced) 2013, 3, (-1)]

(A) $x^2 + y^2 - 6x + 8y + 9 = 0$ (B) $x^2 + y^2 - 6x + 7y + 9 = 0$
(C) $x^2 + y^2 - 6x - 8y + 9 = 0$ (D) $x^2 + y^2 - 6x - 7y + 9 = 0$

CR0145

2. A circle S passes through the point (0, 1) and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then :- [JEE(Advanced)-2014, 3]

(1) radius of S is 8 (B) radius of S is 7
(3) centre of S is (-7, 1) (D) centre is S is (-8, 1)

CR0146

3. Let RS be the diameter of the circle $x^2 + y^2 = 1$, where S is the point (1,0). Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q. The normal to the circle at P intersects a line drawn through Q parallel to RS at point E. then the locus of E passes through the point(s)- [JEE(Advanced)-2016, 4(-2)]

(A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$ (B) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (C) $\left(\frac{1}{3}, -\frac{1}{\sqrt{3}}\right)$ (D) $\left(\frac{1}{4}, -\frac{1}{2}\right)$

CR0147

4. For how many values of p, the circle $x^2 + y^2 + 2x + 4y - p = 0$ and the coordinate axes have exactly three common points ? [JEE(Advanced)-2017, 3]

CR0148

Paragraph "X"

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$.

(There are two question based on Paragraph "X", the question given below is one of them)

5. Let E_1E_2 and F_1F_2 be the chord of S passing through the point $P_0(1, 1)$ and parallel to the x-axis and the y-axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slop -1. Let the tangents to S at E_1 and E_2 meet at E_3 , the tangents of S at F_1 and F_2 meet at F_3 , and the tangents to S at G_1 and G_2 meet at G_3 . Then, the points E_3, F_3 and G_3 lie on the curve [JEE(Advanced)-2018,3(-1)]

(A) $x + y = 4$ (B) $(x - 4)^2 + (y - 4)^2 = 16$
(C) $(x - 4)(y - 4) = 4$ (D) $xy = 4$

CR0149

Paragraph "X"

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$.

(There are two question based on Paragraph "X", the question given below is one of them)

6. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N. Then, the mid-point of the line segment MN must lie on the curve -

[JEE(Advanced)-2018, 3(-1)]

- (A) $(x + y)^2 = 3xy$ (B) $x^{2/3} + y^{2/3} = 2^{4/3}$ (C) $x^2 + y^2 = 2xy$ (D) $x^2 + y^2 = x^2y^2$

CR0150

7. Let T be the line passing through the points P(-2, 7) and Q(2, -5). Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangents to S_1 at P and tangent to S_2 at Q, and also such that S_1 and S_2 touch each other at a point, say, M. Let E_1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1 . Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point R(1, 1) be F_2 . Let E_2 be the set of the mid-points of the line segments in the set F_2 . Then, which of the following statement(s) is (are) TRUE ?

[JEE(Advanced)-2018, 4(-2)]

- (A) The point (-2, 7) lies in E_1
- (B) The point $\left(\frac{4}{5}, \frac{7}{5}\right)$ does NOT lie in E_2
- (C) The point $\left(\frac{1}{2}, 1\right)$ lies in E_2
- (D) The point $\left(0, \frac{3}{2}\right)$ does NOT lie in E_1

CR0151

8. A line $y = mx + 1$ intersects the circle $(x - 3)^2 + (y + 2)^2 = 25$ at the points P and Q. If the midpoint of the line segment PQ has x-coordinate $-\frac{3}{5}$, then which one of the following options is correct ?

[JEE(Advanced)-2019, 3(-1)]

- (1) $6 \leq m < 8$ (2) $2 \leq m < 4$
- (3) $4 \leq m < 6$ (4) $-3 \leq m < -1$

CR0152

9. Let the point B be the reflection of the point A(2, 3) with respect to the line $8x - 6y - 23 = 0$. Let Γ_A and Γ_B be circles of radii 2 and 1 with centres A and B respectively. Let T be a common tangent to the circles Γ_A and Γ_B such that both the circles are on the same side of T. If C is the point of intersection of T and the line passing through A and B, then the length of the line segment AC is _____

[JEE(Advanced)-2019, 3(0)]

CR0153

Answer the following by appropriately matching the lists based on the information given in the paragraph

Let the circles $C_1 : x^2 + y^2 = 9$ and $C_2 : (x-3)^2 + (y-4)^2 = 16$, intersect at the points X and Y. Suppose that another circle $C_3 : (x-h)^2 + (y-k)^2 = r^2$ satisfies the following conditions :

- (i) centre of C_3 is collinear with the centres of C_1 and C_2
- (ii) C_1 and C_2 both lie inside C_3 , and
- (iii) C_3 touches C_1 at M and C_2 at N.

Let the line through X and Y intersect C_3 at Z and W, and let a common tangent of C_1 and C_3 be a tangent to the parabola $x^2 = 8\alpha y$.

There are some expression given in the List-I whose values are given in List-II below :

	List-I	List-II
(I)	$2h + k$	(P) 6
(II)	$\frac{\text{Length of ZW}}{\text{Length of XY}}$	(Q) $\sqrt{6}$
(III)	$\frac{\text{Area of triangle MZN}}{\text{Area of triangle ZMW}}$	(R) $\frac{5}{4}$
(IV)	α	(S) $\frac{21}{5}$
		(T) $2\sqrt{6}$
		(U) $\frac{10}{3}$

10. Which of the following is the only INCORRECT combination ? [JEE(Advanced)-2019, 3(-1)]
 (1) (IV), (S) (2) (IV), (U) (3) (III), (R) (4) (I), (P)

CR0154

11. Which of the following is the only CORRECT combination ? [JEE(Advanced)-2019, 3(-1)]
 (1) (II), (T) (2) (I), (S) (3) (I), (U) (4) (II), (Q)

CR0154

12. Let O be the centre of the circle $x^2 + y^2 = r^2$, where $r > \frac{\sqrt{5}}{2}$. Suppose PQ is a chord of this circle and the equation of the line passing through P and Q is $2x + 4y = 5$. If the centre of the circumcircle of the triangle OPQ lies on the line $x + 2y = 4$, then the value of r is _____ **[JEE(Advanced)-2020]**

CR0159

- 13.** Consider a triangle whose two sides lie on the x-axis and the line $x + y + 1 = 0$. If the orthocenter of is $(1, 1)$, then the equation of the circle passing through the vertices of the triangle is
- [JEE(Advanced)-2021]**

- (A) $x^2 + y^2 - 3x + y = 0$ (B) $x^2 + y^2 + x + 3y = 0$
(C) $x^2 + y^2 + 2y - 1 = 0$ (D) $x^2 + y^2 + x + y = 0$

CR0160

Paragraph for Question 14 and 15

Let $M = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 \leq r^2\}$,

where $r > 0$. Consider the geometric progression $a_n = \frac{1}{2^{n-1}}$, $n = 1, 2, 3, \dots$. Let $S_0 = 0$ and, for $n \geq 1$,

let S_n denote the sum of the first n terms of this progression. For $n \geq 1$, let C_n denote the circle with center $(S_{n-1}, 0)$ and radius a_n , and D_n denote the circle with center (S_{n-1}, S_{n-1}) and radius a_n .

[JEE(Advanced)-2021]

14. Consider M with $r = \frac{1025}{513}$. Let k be the number of all those circles C_n that are inside M . Let l be the maximum possible number of circles among these k circles such that no two circles intersect. Then
 (A) $k + 2l = 22$ (B) $2k + l = 26$ (C) $2k + 3l = 34$ (D) $3k + 2l = 40$

CR0161

15. Consider M with $r = \frac{(2^{199} - 1)\sqrt{2}}{2^{198}}$. The number of all those circles D_n that are inside M is
 (A) 198 (B) 199 (C) 200 (D) 201

CR0162

16. Let ABC be the triangle with $AB = 1$, $AC = 3$ and $\angle BAC = \frac{\pi}{2}$. If a circle of radius $r > 0$ touches the sides AB , AC and also touches internally the circumcircle of the triangle ABC , then the value of r is _____.

[JEE(Advanced)-2022]

CR0189

17. Let G be a circle of radius $R > 0$. Let $G_1, G_2, \dots, G_n$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \dots, G_n$ touches the circle G externally. Also, for $i = 1, 2, \dots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are TRUE?

[JEE(Advanced)-2022]

- (A) If $n = 4$, then $(\sqrt{2} - 1)r < R$ (B) If $n = 5$, then $r < R$
 (C) If $n = 8$, then $(\sqrt{2} - 1)r < R$ (D) If $n = 12$, then $\sqrt{2}(\sqrt{3} + 1)r > R$

CR0190

ANSWER KEY

Do yourself - 1

- Centre $\left(\frac{3}{4}, -\frac{5}{4}\right)$, Radius $\frac{3\sqrt{10}}{4}$
- (i) $x^2 + y^2 + 10x + 12y = 39$; (ii) $x^2 + y^2 - 2ax + 2by = 2ab$
- (i) $(2, 4); \sqrt{61}$; (ii) $\left(\frac{c}{\sqrt{1+m^2}}, \frac{mc}{\sqrt{1+m^2}}\right); c$
- $17(x^2 + y^2) + 2x - 44y = 0$
- (a) $x^2 + y^2 - 6x \pm 6\sqrt{2}y + 9 = 0$; (b) $x^2 + y^2 + 4x - 10y + 4 = 0$; $x^2 + y^2 - 4x - 2y + 4 = 0$
- $x^2 + y^2 + 6x - 2y - 51 = 0$
- $x^2 + y^2 - hx - ky = 0$
- $x = \frac{p}{2}(-1 + \sqrt{2} \cos \theta)$; $y = \frac{p}{2}(-1 + \sqrt{2} \sin \theta)$
- (i) $(11, 16)$, (ii) $(11, 8)$, (iii) $(11, 12)$
- (i) $x^2 + y^2 - ax - by = 0$; (ii) $x^2 + y^2 - 22x - 4y + 25 = 0$; (iii) $x^2 + y^2 - 5x - y + 4 = 0$

Do yourself - 2

- $(1, 2)$ lie inside the circle and the point $(6, 0)$ lies outside the circle
- min = 0, max = 6, power = 0
- square of side 2; $x^2 + y^2 = 1$; $x^2 + y^2 = 2$
- (a) 2; (b) (9, 3)

Do yourself - 3

- $x \cos \alpha + y \sin \alpha = a(1 + \cos \alpha)$
- $4x - 3y + 7 = 0$ & $4x - 3y - 43 = 0$
- $5x + 12y = \pm 26$; $\left(\mp \frac{10}{13}, \mp \frac{24}{13}\right)$
- 1
- (i) $5x - 12y = 152$, (ii) $k = 40$ or -10
- $a^2(x^2 + y^2) = 4x^2y^2$
- 6
- (a) $4x + 3y + 19 = 0$ and $4x + 3y - 31 = 0$;
(b) $12x - 5y + 8 = 0$ and $12x - 5y - 252 = 0$;
(c) $x - \sqrt{3}y \pm 10 = 0$
- $(5, 1)$ & $(-1, 5)$
- $x^2 + y^2 = a^2 + b^2$; $r = \sqrt{a^2 + b^2}$
- $x^2 + y^2 - 2x - 2y + 1 = 0$ OR $x^2 + y^2 - 42x + 38y - 39 = 0$
- $x - y = 0$; $x + 7y = 0$
- $2x - 2y - 3 = 0$
- $4x - 3y - 25 = 0$ OR $3x + 4y - 25 = 0$

Do yourself - 4

- $x + 2y = 1$
- 1

Do yourself - 5

- $4x + 7y + 10 = 0$
- $\frac{405\sqrt{3}}{52}$ sq. units
- (i) $3x - 4y = 21$; $4x + 3y = 3$; (ii) A(0, 1) and B(-1, -6); (iii) 90° , $5(\sqrt{2} \pm 1)$ units
(iv) 25 sq. units, 12.5 sq. units; (v) $x^2 + y^2 + x + 5y - 6$, x-intercept = 5; y-intercept = 7
- $x^2 + y^2 + gx + fy = 0$
- $(-25, 50)$

Do yourself - 6

- $5x - 4y + 26 = 0$
- $x^2 + y^2 + gx + fy = 0$
- $\left(\frac{1}{2}, -\frac{1}{2}\right)$, $x + y = 0$
- $x + y = 2$

Do yourself - 7

- $(x - h)^2 + (y - k)^2 = 2a^2$
- 10
- angle between the tangents = 90°

Do yourself - 8

- $x^2 + y^2 + 4x - 7y + 5 = 0$
- $7x^2 + 7y^2 - 10x - 10y - 12 = 0$
- $2x^2 + 2y^2 - 18x - 22y + 69 = 0$ and $x^2 + y^2 - 2y - 15 = 0$
- $x^2 + y^2 - x - 2y = 0$
- (a) $x^2 + y^2 + 4x - 7y + 5 = 0$, (b) $7(x^2 + y^2) + 19x - 19y - 91 = 0$
- $x^2 + y^2 - 6x + 4y = 0$ OR $x^2 + y^2 + 2x - 8y + 4 = 0$
- $x^2 + y^2 + x - 6y + 3 = 0$
- $\left(2, \frac{23}{3}\right)$

Do yourself - 9

- $(x - 5)^2 + (y - 5)^2 = 25$
- 4
- B
- D
- zero, zero
- $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

Do yourself - 10

- 135° or 45°
- $x + 2y = 2$
- (1, 2)
- B
- (1, 2)
- $4x^2 + 4y^2 + 6x + 10y - 1 = 0$
- $a^{-2} + b^{-2} = c^{-1}$

Do yourself - 11

- 18
- A
- B
- A
- $x^2 + y^2 - 4x - 2y + 3 = 0$
- $x^2 + y^2 + 16x + 14y - 12 = 0$
- $(-4, 4)$; $(-1/2, 1/2)$
- (a) $x^2 + y^2 + 4x - 6y = 0$; $k = 1$; (b) $x^2 + y^2 = 64$
- $x^2 + y^2 - 16x - 18y - 4 = 0$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	A	B	D	A	B	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	B	A	B	A	D	A	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	A	C	A	C	C	C	B	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,B,C,D	A,C,D	A,B,C	C,D	B,C	A,D	A,C	B,D	A,B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C	A,C	A,C,D	A,C,D	A,B,D	A,C	A,C,D	A,C	B,C	B,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	D	C	A	A,B,C,D	A,B	5.00	24.00	2.20	A	
Q.10	A	B	C	D						
	S	R	Q	P						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	10	4	32	19	63	4.00	4.00	3.00	215.00	625.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	1	1	4	2	4	2	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	4	2	36	3	2	9	3	1
Que.	21	22	23	24	25					
Ans.	7	D	16	C	A					

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	B,C	A,C	2	A	D	B,D	2	10.00	1
Que.	11	12	13	14	15	16	17			
Ans.	4	2	B	D	B	0.83 or 0.84	C,D			

Important Notes

